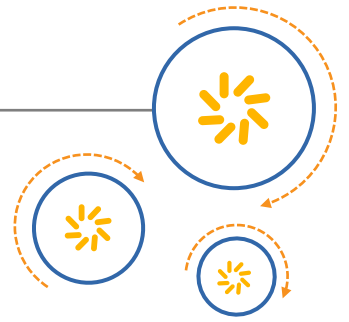




Qualcomm Technologies, Inc.



Controlling Rx Chains on LTE

80-N5220-2 E

November 9, 2017

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Revision history

Revision	Date	Description
A	June 2016	Initial release
B	July 2016	Updated the document title; updated the introduction in chapter 2; updated section 2.1.2.
C	September 2016	Updated the introduction in chapter 2; updated sections 2.1.4, 2.2 and 2.3.
D	February 2017	Updated Section 2.2
E	November 2017	Added Section 2.1

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1 Introduction

1.1 Purpose

This document discusses how to control primary and secondary Rx chains on LTE technology.

1.2 Conventions

Function declarations, function names, type declarations, attributes, and code samples appear in a different font, for example, `#include`.

Code variables appear in angle brackets, for example, `<number>`.

Commands to be entered appear in a different font, for example, **copy a:*. * b:.**

Shading indicates content that has been added or changed in this revision of the document.

1.3 Technical assistance

For assistance or clarification on information in this document, submit a case to Qualcomm Technologies, Inc. (QTI) at <https://createpoint.qti.qualcomm.com/>.

If you do not have access to the CDMATech Support website, register for access or send email to support.cdmatech@qti.qualcomm.com.

2 Rx chains configuration

By default, when LTE calls Rx tune, it requests the Rx driver to turn on both Rx0 and Rx1. Under certain specific test setups, it may be appropriate that one of the two chains (or four chains in HoRxD case) be active and the other inactive.

There are multiple ways to control the primary and diversity path of the User Equipment (UE).

The following methods are used to select single chain, all two chains (Non-HoRxD), or all four chains in 4Rx case (HoRxD) to perform sensitivity measurements.

To verify the command functionality, it is recommended to perform power sweep and sensitivity test.

2.1 Device setup for Rx chain configuration

NOTE: This section was added to this document revision.

Use EFS file `ho_rxd_config` = {1, 0, 0, 0, 50, 55, 0, 0} under `/nv/item_files/modem/lte/ML1/`:

Ensure that the call is connected and that all chains are running (Rx0/1/2/3 in 4Rx mode, and Rx0/1 in 2Rx mode) before triggering any of the following commands. In addition, ensure that you use the commands to enable all chains, before ending the call or switching to another band.

2.2 Configure primary/diversity path

2.2.1 Use EFS file `rx_select` under `/nv/item_files/modem/lte/ML1/`

To turn off a specific Rx chain, place an EFS file with filename **rx_select** in to the User Equipment (UE) Embedded File System (EFS) folders, under the following directory:

`/nv/item_files/modem/lte/ML1/`

The file contains 1 byte, and valid values are as defined in [Table 2-1](#).

Table 2-1 EFS file configuration values

Rx3 (bit 3)	Rx2 (bit2)	Rx1 (bit 1)	Rx0 (bit 0)	EFS values	Summary
0	0	0	1	1	Enables Rx0 only
0	0	1	0	2	Enables Rx1 only
0	0	1	1	3	Enables Rx0 and Rx1 (2Rx mode only)
0	1	0	0	4	Enables Rx2 only
1	0	0	0	8	Enables Rx3 only

Rx3 (bit 3)	Rx2 (bit2)	Rx1 (bit 1)	Rx0 (bit 0)	EFS values	Summary
1	1	1	1	15	Enables all chains

The UE reads the EFS file at power-up, if it exists.

2.2.2 Use the DIAG command

When LTE sends an Rx tuning request to the RF driver, the DIAG command can be sent from the QXDM Professional™ tool (QXDM) to the protocol stack to enable/disable a specific chain. These commands are used to select a single chain or all four chains in the 4Rx Higher-Order Receive Diversity case (HoRxD), and a single chain or all two chains (Non-HoRxD) to perform sensitivity measurements.

NOTE: The DIAG command can only be sent when the RF driver Rx is tuned.

1. The following DIAG command format can be used from the QXDM DIAG command box. This command only works in 2Rx (Non-HoRxD):

```
send_data 75 68 00 64 tool_chain_code
```

Table 2-2 tool_chain_code configuration values

Enable/disable (bit 2)	Rx1 (bit 1)	Rx0 (bit 0)	Values	Summary
0	0	0	0	No action
0	0	1	1	Enables Rx0 only
0	1	0	2	Enables Rx1 only
0	1	1	3	Enables Rx0 and Rx1
1	0	0	4	No action
1	0	1	5	Disables Rx0 only
1	1	0	6	Disables Rx1 only
1	1	1	7	Disables Rx0 and Rx1

2. The following DIAG command format can also be used from the QXDM DIAG command box:

NOTE: The following DIAG command can only be supported starting from TH2.0.1 (with CR 1024285) and it works for both 2Rx (Non-HoRxD) and 4Rx (HoRxD):

```
send_data 75 68 9 64 <8-bit config_number>
```

This command (Command ID: 0x4000) is routed to the subsystem DIAG_SUB_SYS_LTE (68). The tool_chain_code is a 1-byte value. See the following byte description:

Command description

This DIAG command enables or disables a specific Rx chain while the UE is in LTE mode and camped on a specific system.

Command payload

The payload to this DIAG command is 1 byte.

- Bit 7 to bit 4 – CC ID
- Bit 3 – enabling or disabling Rx3
 - 0 – Disable Rx3
 - 1 – Enable Rx3
- Bit 2 – enabling or disabling Rx2
 - 0 – Disable Rx2
 - 1 – Enable Rx2
- Bit 1 – enabling or disabling Rx1
 - 0 – Disable Rx1
 - 1 – Enable Rx1
- Bit 0 – enabling or disabling Rx0
 - 0 – Disable Rx0
 - 1 – Enable Rx0
- **cc_id**: carrier number: PCell = 0, SCell 1 = 1, SCell 2 = 2, and so on.

Table 2-3 is a usage summary for bit 3 to bit 0.

Table 2-3 config_number configuration values

Rx3 (bit 3)	Rx2 (bit2)	Rx1 (bit 1)	Rx0 (bit 0)	config_number	Summary
0	0	0	1	1	Enables Rx0 only
0	0	1	0	2	Enables Rx1 only
0	0	1	1	3	Enables Rx0 and Rx1
0	1	0	0	4	Enables Rx2 only
1	0	0	0	8	Enables Rx3 only
1	1	1	1	15	Enables all chains

NOTE: Only the configuration numbers listed in Table 2-3 can be used.

2.2.3 Use the AT command

To add support for configuring the primary or diversity chain as per requirements using an AT command, licensees should use their own implementation of the AT command and should call the newly added CM API (see the following). (CR 1023340)

For more details on ATCoP support, see *Comprehensive AT Command Set in AMSS Software* (80-VR432-1).

`cmapi_set_lte_common_rx_diversity (cc_id, selection_mask)`

- **cc_id**: carrier number: PCell = 0, SCell 1 = 1, SCell 2 = 2, etc.
- **selection_mask**: cannot be 0 and only one chain or all chains must be ON

Table 2-4 selection_mask configuration values

Rx3 (bit 3)	Rx2 (bit2)	Rx1 (bit 1)	Rx0 (bit 0)	EFS values	Summary
0	0	0	1	1	Enables Rx0 only
0	0	1	0	2	Enables Rx1 only
0	0	1	1	3	Enables Rx0 and Rx1
0	1	0	0	4	Enables Rx2 only
1	0	0	0	8	Enables Rx3 only
1	1	1	1	15	Enables all chains

2.2.4 Use the FTM command

To use FTM Rx Select command in 4Rx (HoRxD), enable 4Rx using the following FTM NS commands:

```
QLIB_FTM_LTE_NS_START()
QLIB_FTM_LTE_NS_ACQ(1, 300)
QLIB_FTM_LTE_NS_START_DP_ENH(43690, 3, 24, 1, 2, 4)
QLIB_DIAG_LTE_RX_CHAIN_SELECTION_COMMAND(0x4)
```

To use FTM Rx Select command in 2Rx (non-HoRxD), enable 2Rx using the following FTM NS commands:

```
QLIB_FTM_LTE_NS_START()
QLIB_FTM_LTE_NS_ACQ(1, 300)
QLIB_FTM_LTE_NS_START_DP(43690, 3, 24, 1, 2)
```

After enabling the Rx chains, the FTM command can be used to select a single Rx chain.

The details of the command are listed in [Table 2-5](#) and [Table 2-6](#).

For more information, refer to *FTM API - LTE Nonsignaling Commands ICD* (80-N2360-1).

Table 2-5 FTM_LTE_NS_RX_SELECT_CHAIN_REQ request packet

Field	Type/length (bytes)	Description
CMD_CODE	Unsigned/1 byte	Command ID – Set CMD_CODE to 75
SUB_SYS_ID	Unsigned/1 byte	Subsystem ID – FTM ID is 11
RF_MODE_ID	Unsigned/2 bytes	▪ RF Mode ID – FTM LTE NS is 30
RF_SUB_CMD	Unsigned/2 bytes	▪ Name – FTM_LTE_NS_RX_SELECT_CHAIN_REQ ▪ ID – 541
CARRIER_0_CHAIN_MASK	Unsigned/2 bytes	Carrier 0 Rx chain mask using <code>rfte_mc_carrier_type</code> as defined in the following note.
CARRIER_1_CHAIN_MASK	Unsigned/2 bytes	Carrier 1 Rx chain mask using <code>rfte_mc_carrier_type</code> as defined in the following note.

Field	Type/length (bytes)	Description
CARRIER_2_CHAIN_MASK	Unsigned/2 bytes	Carrier 2 Rx chain mask using rflte_mc_carrier_type as defined in the following note.
CARRIER_3_CHAIN_MASK	Unsigned/2 bytes	Carrier 3 Rx chain mask using rflte_mc_carrier_type as defined in the following note.
CARRIER_4_CHAIN_MASK	Unsigned/2 bytes	Carrier 4 Rx chain mask using rflte_mc_carrier_type as defined in the following note.

NOTE: rflte_mc_carrier_type: PRx = 0; DRX = 1; drx2 = 2; DRX 3 = 3

Table 2-6 FTM_LTE_NS_RX_SELECT_CHAIN_REQ response packet

Field	Length (bytes)	Description
CMD_CODE	Unsigned/1 byte	Command ID – Set CMD_CODE to 75
SUB_SYS_ID	Unsigned/1 byte	Subsystem ID – FTM ID is 11
RF_MODE_ID	Unsigned/2 bytes	RF Mode ID – FTM LTE NS is 30
RF_SUB_CMD	Unsigned/2 bytes	- Name – FTM_LTE_NS_RX_SELECT_CHAIN_REQ - ID – 541
STATUS	Unsigned/4 bytes	0 = Success 1 = Failure 2 = Not Allowed 12 = In progress 13 = Already done 16 = Invalid Argument 20 = Bad data 55 = Busy

2.3 Recommendations

In the MSM8998 chipset, the 0xB193 log packet can be used to check whether the QXDM command is working on software. The RSSI, RSRP, RSRQ, and FTL SNR for disabled chains show the floored values, as shown in the following example:

Rx0 Only:

New logging (With CR 1098507 and CR 1103145 and QCAT 6.30 ver 39.3)

```

1980 Jan 6 00:44:32.948 [BE] 0xB193 LTE ML1 Serving Cell Meas Response
Version = 1
Number of SubPackets = 1
SubPacket ID = 25
Serving Cell Measurement Result
  Version = 35
  SubPacket Size = 508 bytes
  E-ARFCN = 1575

```

```

Num of Cell          = 1
Valid Rx             = RX0
...
Inst RSRP Rx[0]      = -115.38 dBm
True Inst RSRP Rx[0] = -115.38 dBm
Inst RSRP Rx[1]      = NA
True Inst RSRP Rx[1] = -180.00 dBm
...
Inst RSRQ Rx[0]      = -10.50 dB
True Inst RSRQ Rx[0] = -10.50 dB
Inst RSRQ Rx[1]      = NA
True Inst RSRQ Rx[1] = -30.00 dB
...
Inst RSSI Rx[0]      = -87.88 dBm
Inst RSSI Rx[1]      = NA
...
FTL SNR Rx[0]        = 13.30 dB
FTL SNR Rx[1]        = NA

RS SNR Rx[0]         = 13.10 dB
RS SNR Rx[1]         = -20.00 dB

```

Old logging

```

2016 Jun 30 21:09:55.441 [02] 0xB193 LTE ML1 Idle Serving Cell Meas
Response
Version          = 1
Number of SubPackets = 1
SubPacket ID     = 25
Serving Cell Measurement Result
  Version = 22
  SubPacket Size = 492 bytes
  E-ARFCN          = 40620
  Num of Cell      = 1
  horxd_mode = 1
...
Inst RSRP Rx[0]      = -100.31 dBm
True Inst RSRP Rx[0] = -100.31 dBm
Inst RSRP Rx[1]      = -180.00 dBm
True Inst RSRP Rx[1] = -180.00 dBm
Inst RSRP Rx[2]      = -180.00 dBm
Inst RSRP Rx[3]      = -180.00 dBm
...
Inst RSRQ Rx[0]      = -10.00 dB
True Inst RSRQ Rx[0] = -10.00 dB
Inst RSRQ Rx[1]      = -30.00 dB

```

```

True Inst RSRQ Rx[1]      = -30.00 dB
Inst RSRQ Rx[2]          = -30.00 dB
Inst RSRQ Rx[3]          = -30.00 dB
...
Inst RSSI Rx[0]           = -70.31 dBm
Inst RSSI Rx[1]           = -110.00 dBm
Inst RSSI Rx[2]           = -110.00 dBm
Inst RSSI Rx[3]           = -110.00 dBm
...
FTL SNR Rx[0]             = 20.60 dB
FTL SNR Rx[1]             = -20.00 dB
FTL SNR Rx[2]             = -20.00 dB
FTL SNR Rx[3]             = -20.00 dB

```

In addition to checking the B193 packet, it is recommended to perform power sweep and check sensitivity to verify if the QXDM command is functioning.

For commercial devices, it is advised that all the chains are enabled, that is, default configuration.

2.4 Related CRs

[Table 2-7](#) lists the related CRs for the tool chain function.

Table 2-7 Related CRs

CR number	CR title
992860	THOR 2.0.1 toolchain updates for 4Rx (ML1 UTILS)
992858	THOR 2.0.1 toolchain updates for 4Rx (firmware)
1024285	EFS/QXDM command to disable PRx/DRx does not work
1023340	[HoRxD] Crash: cmapi_change_rx_diversity(), cmapi_lte_cc_horxd_rx_diversity()
1025445	FTM_LTE_NS_RX_SELECT_CHAIN_REQ in TH2.0.1
1027175	[RFA]: FTM_LTE_NS_RX_SELECT_CHAIN_REQ in TH.2.0.1
1027187	[RF_FACTORY]: FTM_LTE_NS_RX_SELECT_CHAIN_REQ in TH.2.0.1
1009082	[TH.2.0.1] HoRxD Does not work in FTM Non-signaling mode
1031046*	[HoRxD] New Request for Rx Tool chain - RSRP/SNR for disabled chains
1098507*	Log fields in B193 need to be logged per physical antenna
1103145*	8998 ARD, 0xB193 does not show Rx2 and Rx3 values in Best2of4 or HoRxd mode

*CR 1031046, 1098507, 1103145 is only supported on the MSM8998 and MDM9x50 chipsets.

A References

A.1 Related documents

Title	Number
Qualcomm Technologies, Inc.	
<i>Comprehensive AT Command Set in AMSS Software</i>	80-VR432-1
<i>FTM API - LTE Nonsignaling Commands ICD</i>	80-N2360-1

A.2 Acronyms and terms

Acronym or term	Definition
EFS	Embedded file system
HORXD	Higher order receive diversity
QXDM	Qualcomm® extensible diagnostic monitor
UE	User equipment